MIL-S-36586A
18 August 1976
SUPERSEDING
MIL-S-36586
20 October 1969

MILITARY SPECIFICATION

STERILIZATION TEST STRIP SET, BACTERIAL SPORE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers bacterial spore test strip sets, suitable for use as biological indexes in determining the efficiency of sterilization processes by bacteriological culture test. (see 6.1)

1.2 Classification. Sterilization test strip sets shall be either of the following types, as specified (see 3.3.2):

Type A	Bacillus subtilis, var. niger
	Bacillus stearothermophilus
Type B	Bacillus subtilis, var. globigii
	Bacillus stearothermophilus

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping Fiberboard.
PPP-B-676 - Boxes, Set-Up.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 S. 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6530

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MILITARY

MIL-L-10547 - Liners, Case and Sheet, Overwrap, Water-Vaporproof, or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

THE WILLIAMS AND WILKINS COMPANY, A DIVISION OF WAVERLY PRESS, INC.

Bergey's Manual of Determinative Bacteriology.

(Application for copies should be addressed to the Williams and Wilkins Co., A Division of Waverly Press Inc., 428 E. Preston St., Baltimore, MD 21202.)

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI) STANDARD

T-410 - Basis Weight of Paper and Paperboard.

(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, 1 Dunwoody Park, Atlanta, GA 30341.)

3. REQUIREMENTS

* 3.1 First article. When specified in the procurement document, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.1.2.1 and 6.2).

3.2 Material.

3.2.1 Filter paper. Filter paper shall be of a grade and absorbency such that it can be impregnated with bacterial spore populations specified herein.

3.2.2 Glassine paper. Glassine paper shall be sterilization blue with a basis weight of 30 pounds per 500 sheets, 24 by 36 inches, when tested as specified in 4.4.2.

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3.2.3 Envelope paper. Envelope paper used in the procedure envelope shall be white, with a basis weight of 20 pounds per 500 sheets, 24 by 36 inches, when tested as specified in 4.4.2.

3.3. Construction. Each sterilization test strip set shall consist of a procedure envelope containing three test strips, each contained in a sealed glassine envelope.

3.3.1 Test strip. Each test strip shall consist of an absorptive strip of filter paper not less than 1/4 inch wide by 1-3/8 inches long, impregnated with a mixed population of Bacillus stearothermophilus and bacillus subtilus (globigii or var. niger) viable dry bacterial spores. The strips shall be inoculated with spore populations adjusted to give predetermined resistance (survival and kill) to saturated steam under pressure, dry heat and to ethylene oxide, when tested as specified in 4.4.3. Spore monitors shall be free of extraneous microbial contamination such as yeast and mold. Bacterial spore identification shall be as indicated in paragraph 4.4.1.

3.3.2 Glassine envelope. Three sterilizable glassine envelopes, measuring 1 inch wide by at least 2 inches long, shall be provided for containing one test monitor each. Two envelopes shall bear the following in black ink as applicable:

Type A

BACTERIAL SPORE
STERILIZATION MONITOR

Bacillus subtilus var. niger and Bacillus stearothermophilus.
Test Piece
Manufacturer's Name or Registered Trademark
Lot Number

Type B

BACTERIAL TEST STRIP
FOR STERILIZATION PROCEDURES

Gas, Dry Heat or Steam (250° or 270°F.) (121° or 132°C.)
Bacillus stearothermophilus
Bacillus subtilis (globigii)
Manufacturer's Name or Registered Trademark
Lot Number

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As applicable, one envelope shall bear the following in red ink:

Type A

BACTERIAL SPORE
STERILIZATION MONITOR
Bacillus subtilis var. niger and
Bacillus stearothermophilus.
-CONTROL PIECE-
Manufacturer's Name or Registered Trademark
Lot Number

Type B

VIABILITY CONTROL STRIP
Bacillus stearothermophilus
Bacillus subtilis (globigii)

Manufacturer's Name or Registered Trademark
Lot Number

* As an alternate to imprinting the lot number on the glassine envelope, it may be imprinted on each spore strip. The adhesive sealer used to seal the glassine envelope shall in no way impair the performance of the impregnated filter paper strips.

3.3.3 Procedure envelope. A double-pocket, white paper, procedure envelope shall contain two test strips (printed in black) placed in the test section of the procedure envelope, the flap of which shall be unsealed; and one control test strip (printed in red) placed in the control section of the procedure envelope, the flap of which shall be sealed.

As applicable, the following minimum information shall be printed on front of the envelope:

Type A

INSTRUCTIONS FOR USING STERILIZATION MONITORS

ORGANISMS: BACILLUS SUBTILIS var. NIGER and BACILLUS STEAROTHERMOPHILUS

This envelope contains two test monitors
SUPERVISOR
Instructions for Sterilization

This envelope contains one control monitor
BACTERIOLOGIST
Instructions for culturing

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1. Include the two test strips from this pocket within pack or materials to be sterilized, and locate in the most difficult area to be sterilized by the sterilization agent used (see procedure instructions for additional details).
2. Fill in data required on other side of this envelope.
3. After sterilization, send monitors in this envelope to Bacteriology Laboratory for testing.

Lot Number Expiration Date:

1. Aseptically open the glassine envelopes, place control strips into individual tubes of trypticase soy broth. Identify all tubes.
2. Incubate seven days at specified temperature.
   a. 55°C. (131°F.) for steam sterilization.
   b. 32° - 37°C. (90° - 99°F.) for Ethylene Oxide, dry heat, and radiation sterilization.
   c. For other applications see instruction sheet.
3. Observe tubes daily for growth. Growth should occur in unsterilized control within 48 hours.
4. Record results on this envelope and return to sending department for filing.

* Type B

PROCEDURE FOR DETERMINING EFFICIENCY OF GAS, STEAM, OR DRY HEAT STERILIZATION

SUPERVISOR

Contents: Two Test Strips
Organisms: B. stearothermophilus &
B. subtilis

1. Place two test strips from this side of envelope and locate in the most difficult area to be sterilized.
2. Sterilize according to normal procedure.
3. Fill in reverse side of this envelope.
4. Take load from sterilizer. Remove strips.
5. Put both strips into this envelope, seal and deliver to Bacteriology Laboratory.
6. Store at ambient conditions remote from ethylene oxide.
7. After expiration date, autoclave any remaining test sets and discard.

Lot No. Expiration Date

BACTERIOLOGY LABORATORY

Contents: One Control Strip
Organisms: B. stearothermophilus &
B. subtilis

1. Aseptically cut open glassine envelope. Culture both test strips and control strip in TRYPTICASE SOY BROTH. B.B.L. #11768.
2. Incubate tubes 7 days at proper temperature 55°C. (131°F.) if strips were exposed to steam. 37°C. (99°F.) if strips exposed to gas or dry heat. (See sterilization test data on other side of envelope for sterilizing agent used).
3. Observe tubes daily for bacterial growth.
4. Record results on reverse side of this envelope. Return envelope to Supervisor initiating test.
5. When test is completed, autoclave position cultures.

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The back of the envelope shall be imprinted with the following minimum information, as applicable:

Type A

BACTERIOLOGY CULTURE REPORT

Date Monitors were Cultured _____
Control Monitor: Observed growth in ___ days
Monitor #1 ☐ No growth
☐ Growth observed in ___ days
Monitor #2 ☐ No growth
☐ Growth observed in ___ days
Incubated at ☐ 32° - 37°C. (90°-99°F.)
☐ 55°C. (131°F.)
☐ Observed growth was checked microscopically and gram stained.

Signature _____
Date _____

STERILIZATION REPORT

Institution _____
Date of Cycle _____
Sterilizer _____
Cycle # _____
Location of Monitors #1 _____
#2 _____
Sterilization Conditions
☐ STEAM ☐ ETHYLENE OXIDE
☐ Dry Heat ☐ Other
Temperature _____
Exposure period _____
Signature _____
Department _____

Type B

STERILITY TEST REPORT
FROM BACTERIOLOGY LABORATORY

To _____ (Supervisor)
Department _____
Date strips were cultured _____
Temperature at which strips were incubated:
37°C.(99°F.)☐ 55°C.(131°F.)☐
Results of culture tests (check one)
Both strips NEGATIVE ☐ POSITIVE ☐
One strip NEGATIVE the other POSITIVE
Control Strip results:
POSITIVE ☐ NEGATIVE ☐
☐ Observed growth was checked microscopically and gram stained.
Should test be repeated: NO ☐ YES ☐
Signature _____
Date _____

STERILIZATION TEST DATA

Hospital _____
Date of test _____
Sterilization Agent Steam ☐
Gas ☐
Dry Heat ☐
Location of sterilizer _____
Type of load _____
Sterilizing conditions:

Test conducted by _____
Department _____

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The word "Supervisor" shall be printed on the flap for sealing the pocket containing the Sterilization Report. The word "Bacteriologist" or "Bacteriology Laboratory" shall be printed on the flap for sealing the Bacteriology Culture Report.

3.3.4 Potency period. The test sets shall have an 18-month potency period. Not less than 15 months of the potency period shall remain at time of delivery to the Government. Each procedure envelope shall be stamped or otherwise permanently imprinted with an expiration date. The manufacturer shall retain samples of each lot for the potency period.

3.4 Workmanship. Workmanship shall be first class throughout. Test strip sets shall be free from defects which detract from their appearance or may impair their serviceability.

* 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection.
b. Quality conformance inspection.

4.1.2.1.1 First article inspection. First article inspection shall be performed on the sets when a first article sample is required (see 3.1). This inspection shall include examination of 4.3 and tests of 4.4.

4.1.2.1.2 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.3 and tests of 4.4.

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4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling. The unit of product for inspection purposes shall be an individual test strip set.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105.

TABLE I. Sampling for tests.

Component (unit of product)	Characteristic	Requirement	Test procedure	Inspection level 1/	AQL
Test strip set	Basis weight	3.2.2 & 3.2.3	4.4.2		
	Resistance to saturated steam under pressure	3.3.1	4.4.3.1	12 Sets	2/
	Spore identification	3.3.1	4.4.1	1 Set	
	Dry Heat	3.3.1	4.4.3.2	12 Sets	2/
	Ethylene oxide	3.3.1	4.4.3.3	12 Sets	2/

1/ For each lot.
2/ The acceptance number shall be zero regardless of lot size. Not less than three samples shall be tested.

4.3 Examination. The test strip sets shall be examined to determine compliance with all requirements contained in this specification.

4.4 Tests. Tests shall be conducted, as necessary, to determine compliance with specification requirements.

4.4.1 Bacterial spore identification. Bacterial spore identification shall be made in accordance with Bergey s Manual of Determinative Bacteriology. In addition, cultures shall be observed to determine freedom from extraneous microbial contamination.

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4.4.2 Basis weight. Test for basis weight of paper shall be conducted in accordance with method T 410 of TAPPI.

4.4.3 Resistance.

4.4.3.1 Saturated steam under pressure. Test strips shall be tested in an autoclave for the periods and at the temperatures specified in table II.

TABLE II. Saturated steam under pressure.

	Temperature Type A and Type B	Time (Minutes) Type A	Time (Minutes) Type B	Incubation Type A and Type B
Survival	252°F. + 2° (122°C. + 1°)	5	5	55°C. (131°F.)
	272°F. ± 2° (133°C. ± 1°)	1/2	1/3	55°C. (131°F.)
Kill	252°F. + 2° (122°C. + 1°)	15	15	55°C. (131°F.)
	272°F. ± 2° (133°C. ± 1°)	3	2	55°C. (131°F.)

4.4.3.2 Dry heat. Test strips shall be tested in a hot air oven for the periods and at the temperature specified in table III.

TABLE III. Dry heat.

	Temperature Type A and Type B	Time (Minutes) Type A	Time (Minutes) Type B	Incubation Type A and Type B
Survival	250°F. + 2° (121°C. + 1°)	60	30	37°C. (99°F.)
Kill	320° to 338°F. (160° to 170°C.).	240	120	37°C. (99°F.)

4.4.3.3 Ethylene oxide. Test strips shall be tested in gaseous ethylene oxide under the conditions specified in table IV.

TABLE IV. Ethylene Oxide.

	Temperature Type A and Type B	Ethylene Oxide Concentration (Milligrams per Liter) Type A	Ethylene Oxide Concentration (Milligrams per Liter) Type B	Time (Minutes) Type A	Time (Minutes) Type B
Survival	130°F. (54°C.)	1200	600	5	15
Kill	130°F. (54°C.)	1200	600	30	120

Relative Humidity: 60% + 20%
Incubation: Type A and Type B 37°C. (99°F.)

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4.5 Packaging inspection. The inspection of the unit, intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

* 5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One package containing 25 procedure envelopes as specified, constitutes one unit of issue.

5.1.1.1 Level A.

5.1.1.1.1 Unit packaging. Each unit of issue shall be packaged in a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676 or a plastic box of appropriate size with a hinged cover. Closure of box shall be as specified in the appendix of the applicable box specification.

5.1.1.1.2 Intermediate packaging. Twelve (12) units of issue shall be packaged in a box of appropriate size constructed in accordance with PPP-B-566, PPP-B-676 or PPP-B-636, type CF, class domestic. Closure of box shall be as specified in the appendix of the applicable box specification.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

5.2.1 Level A. Twenty-four (24) units of issue shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 3; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in the appendix of PPP-B-636.

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5.2.2 Level B. Twenty-four (24) units of issue, shall be packaged in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking. Each unit and intermediate package and exterior container shall be marked as specified in MIL-STD-129. In addition, type I shelf life marking as specified in MIL-STD-129 shall be shown.

6. NOTES

6.1 Intended use. For use in determining the efficiency of any sterilization process; steam at 250°F. and 270°F., (121°C. and 132°C.) gas or dry heat by bacteriological culture test.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
c. When a first article is required (see 3.1, 4.1.2.1 and 6.3).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The contract or purchase order specifies the quantity, procedure for delivery and other specific instructions for inspection and approval of the first article.

6.4 This specification covers the following item appearing in the Federal Supply Catalog:

NSN	Item Identification
6530-00-477-6720	STERILIZATION TEST STRIP SET, BACTERIAL SPORE, Bacillus Subtilis and Bacillus Stearothermophilus, 25s

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6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DSA-DM

Project No. 6530-1398

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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